A Zero-Dimensional Frobenius Obstruction for Periodic Schemes of an Area-Preserving Hénon Family

Anonymous Authors

August 6, 2026

Abstract

We study the fixed-period arithmetic of the area-preserving Hénon family $H_a(q,p) = (1 - aq^2 - p, q)$ while keeping chronological iteration and Frobenius extension degree as independent variables. The periodic-point scheme of H_a^n admits a cyclic recurrence presentation whose equations form a monic Gröbner basis over $\mathbb{Z}[a, a^{-1}]$. It is therefore finite flat of rank 2^n. This exact structure also gives an obstruction: at any finite reduced fiber, Frobenius acts only as a permutation of a zero-dimensional set. The fixed-n local zeta is consequently a finite permutation determinant, all of its eigenvalues are roots of unity, and ordinary point counts ignore nilpotents and local intersection multiplicities. An exact certificate with 36 finite-field cells, eight unit tests, and an independent field implementation verifies the good-prime, ramified, and degree-drop cases. A matched reversible two-cycle control further proves that rectangular point counts do not determine a joint Frobenius/dynamical action even among reversible finite controls. Finally, a promising period-five sextic with Galois group S_6 is shown, coefficient by coefficient, to coincide with a published Hénon orbital polynomial. Thus fixed-period Frobenius rationality does not furnish a distinguished Hilbert–Pólya mechanism; any surviving arithmetic program must use genuinely joint actions or positive-dimensional parameter geometry.

1 Introduction

Separating arithmetic time from dynamical time is necessary when a map is reduced modulo primes, but that separation does not by itself create a new zeta mechanism. This note makes the limitation explicit for the conservative quadratic recurrence

$$H_a(q,p) = (1 - aq^2 - p, q), \qquad q_{i+1} + q_{i-1} + aq_i^2 - 1 = 0. \tag{1}$$

The map has Jacobian determinant one and the rational reversor $R(q,p) = (p,q)$, with $RH_aR = H_a^{-1}$. It is a standard area-preserving Hénon normalization [5, 7].

For an integer a, a prime p, a Frobenius extension degree r, and a chronological period n, consider

$$X_{a,n} = \mathrm{Fix}(H_a^n), \qquad N_{a,p}(r,n) = \#X_{a,n}(\mathbb{F}_{p^r}). \tag{2}$$

The indices in (2) have different meanings: n iterates the Hénon map, whereas r iterates Frobenius. A previous finite-field experiment that identified these clocks had no cohomological justification. The natural repair was to hold n fixed and construct the Hasse–Weil series in r.

The repair is mathematically coherent but too weak. We prove that the universal fixed scheme $\mathcal{X}_n$ is finite flat of rank 2^n over $\mathbb{Z}[a, a^{-1}]$. At a reduced finite-field fiber its entire r-dependence is the cycle type of one finite Frobenius permutation. Hence the desired trace decomposition and rational local zeta are automatic zero-dimensional facts, not anomalous evidence. Nonreduced fibers do not help because ordinary rational points and ℓ-adic cohomology see only the reduction.

The obstruction has four concrete parts.

(i) We give a scheme-theoretic cyclic presentation and a monic Gröbner basis proof of finite flatness with rank exactly 2^n.

(ii) We distinguish coefficient degree-goodness from ètaleness and show that direct reduction at $p \mid a$, outside the inverted base, can make $\mathrm{Fix}(H_a^4)$ positive-dimensional.

(iii) We prove that the fixed-n local factor is a root-of-unity permutation determinant and that $N(r,n)$ loses the relative Frobenius/Hénon phase.

(iv) We reproduce the first tempting period-five Galois signal and identify an exact collision with the Endler–Gallas sextic [4].

The conclusion is deliberately scoped. It rejects fixed-n recurrence discovery as a Hilbert–Pólya route. It does not exclude arithmetic structure in higher-period Galois towers or in positive-dimensional parameter quotients.

2 Setup and source boundaries

Polynomial automorphisms. Friedland and Milnor prove that a cyclically reduced complex polynomial automorphism of degree d has d fixed points counted with local multiplicity; its n-th iterate has total multiplicity d^n [5, Theorem 3.1]. Their generic distinctness lemma is followed by explicit collision examples. Our integral finite-flat statement agrees with this complex count, but its proof is independent and works before choosing a field.

Frobenius zeta functions. Grothendieck's trace formula expresses the zeta function of a finite-type scheme over a finite field through Frobenius on compactly supported ℓ-adic cohomology [6]. In dimension zero the theorem reduces to elementary permutation linear algebra. This reduction, rather than general rationality, is the main obstruction below.

Formal and primitive period. Hutz develops dynatomic cycles for morphisms of nonsingular projective varieties and records the characteristic-dependent difference between formal and primitive period [8]. A Hénon map extends birationally to $\mathbb{P}^2$, not as the projective morphism assumed there. We therefore use the affine fixed scheme directly and invoke set-level Möbius inversion only on reduced fibers. Diller–Favre's surface theory supplies the corresponding compactification caution [2].

Reversibility and higher-rank actions. Reversibility imposes exact finite-field cycle identities [10]; it must be matched by controls rather than interpreted as an arithmetic anomaly. Lind's $\mathbb{Z}^d$ zeta uses common fixed counts for every finite-index subgroup [9]. The table $N(r,n)$ samples only rectangular subgroups of the commuting Frobenius–Hénon action. Moreover, $\mathbb{A}^2(\overline{\mathbb{F}}_p)$ is not the compact metric phase space in Lind's analytic hypotheses. Walton's finite-field periodic zeta gives another warning: for an automorphism, every point of each finite phase space is periodic, so the union-over-all-periods count forgets the map [11]. Walton's Definition 4.6 also directly precedes our twisted counts by using $\#\,\mathrm{Fix}(gF_q^r)$ and character averages.

Low-period arithmetic. Exact orbital polynomials for Hénon maps have a substantial history. Endler–Gallas treat period four arithmetically [3], and their 2006 paper already gives the period-five Z sextic used below, its discriminant, and symmetric Galois group [4]. Brison–Gallas later publish transformations between the companion sextics [1]. These sources are a novelty firewall for the calculation in Section 5.

3 Finite-flat periodic schemes

Let $B = \mathbb{Z}[A, A^{-1}]$ and let H_A denote (1) over B.

Theorem 3.1 (Cyclic presentation and rank)**.** *For every $n \geq 1$, the fixed scheme $\mathcal{X}_n = \mathrm{Fix}(H_A^n)$ is scheme-theoretically isomorphic to*

$$\operatorname{Spec} B[x_0, \ldots, x_{n-1}]/(f_0, \ldots, f_{n-1}), \qquad f_i = Ax_i^2 - 1 + \sum_{j \in \{i-1, i+1\}} x_j, \tag{3}$$

where neighbors are counted as a multiset modulo n. The morphism $\mathcal{X}_n \to \operatorname{Spec} B$ is finite flat of rank 2^n.

Proof. Writing $H_A^i(x_0, x_{-1}) = (x_i, x_{i-1})$ gives the recurrence in (1). The scheme map sends (q, p) to $x_i = \pi_1 H_A^i(q, p)$; its inverse sends a cyclic tuple to (x_0, x_{n-1}). The recurrence verifies both compositions on coordinate rings. For $n = 1$ the inverse is $x_0 \mapsto (x_0, x_0)$, and for $n = 2$ it is $(x_0, x_1) \mapsto (x_0, x_1)$. In particular, the multiset convention gives

$$n = 1: \quad Ax_0^2 + 2x_0 - 1,$$

and

$$n = 2: \quad Ax_0^2 + 2x_1 - 1, \qquad Ax_1^2 + 2x_0 - 1.$$

After division by the unit A, every f_i is monic with leading monomial x_i^2 under any degree-compatible order. These monomials are pairwise coprime, so Buchberger's product criterion applies over B. The standard monomials are $\prod_i x_i^{e_i}$ with $e_i \in \{0, 1\}$, giving a free B-basis of size 2^n. $\square$

The theorem separates total scheme length from geometric support size. The latter equals 2^n only at a reduced fiber.

Proposition 3.2 (Degree-drop primes)**.** *For an integer $a \neq 0$, every prime $p \nmid a$ has fiber scheme length 2^n. A prime $p \mid a$ is not a point of $\operatorname{Spec} \mathbb{Z}[A, A^{-1}]$. Direct reduction of the original uninverted $\mathbb{Z}[A]$-family instead gives*

$$H_0(q, p) = (1 - p, q), \qquad H_0^4 = I.$$

In particular, $\mathrm{Fix}(H_0^n) = \mathbb{A}^2$ whenever $4 \mid n$.

Thus 'degree-good' means $p \nmid a$; it does not mean ètale. Let J_n be the Jacobian of the cyclic equations and let M_n be the derivative monodromy of the Hénon orbit. Linearizing the scalar recurrence identifies $\ker J_n$ with $\ker(I - M_n)$: a cyclic tangent vector satisfies

$$\begin{pmatrix} v_{i+1} \\ v_i \end{pmatrix} = DH_a(x_i, x_{i-1}) \begin{pmatrix} v_i \\ v_{i-1} \end{pmatrix}.$$

Thus a degree-good fiber is ètale precisely when no geometric periodic point has multiplier one. Every degree-good characteristic-two fiber ($a \neq 0$) is non-ètale, since then

$$DH_a = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and every power has eigenvalue one. The qualifier is necessary: at $a = 0$ some fixed schemes are empty, hence ètale, whereas $\mathrm{Fix}(H_0^4) = \mathbb{A}^2$.

For later use, exact norm calculations at the first two periods give

$$D_{a,1} = -4(a + 1), \qquad D_{a,2} = 2^8(a + 1)(a - 3)^3. \tag{4}$$

4 The zero-dimensional Frobenius obstruction

Fix a degree-good fiber over $\mathbb{F}_p$ and write

$$S_{a,p,n} = X_{a,n}(\overline{\mathbb{F}}_p)_{\mathrm{red}}.$$

Theorem 4.1 (Permutation collapse)**.** *For every fixed n and every $r \geq 1$,*

$$N_{a,p}(r,n) = \operatorname{Tr}\big(\mathrm{Frob}_p^r \mid \mathbb{Q}_\ell[S_{a,p,n}]\big)\,. \tag{5}$$

If c_d denotes the number of Frobenius orbits of length d, then

$$Z_{a,p,n}(u) = \exp\left(\sum_{r\geq 1} N_{a,p}(r,n)\frac{u^r}{r}\right) = \prod_d (1-u^d)^{-c_d} = \det(I - u\,\mathrm{Frob}_p \mid \mathbb{Q}_\ell[S_{a,p,n}])^{-1}. \tag{6}$$

Every Frobenius eigenvalue in (6) is a root of unity, and ordinary point counts are unchanged if $X_{a,n}$ is replaced by its reduction.

Proof. Frobenius permutes the finite set $S_{a,p,n}$. The trace of its r-th power counts fixed basis vectors, which are exactly the $\mathbb{F}_{p^r}$-points. Summing the contribution of one cycle of length d gives $\exp(\sum_{d|r} du^r/r) = (1-u^d)^{-1}$. Field-valued points factor through the reduction, so nilpotents do not change either side. □

The theorem applies to every finite zero-dimensional fiber, reduced or not, and mechanically supplies periodicity in r, a finite linear recurrence, sealed-r validation, and local rationality for every finite scheme control. Nonreduced points can be recorded separately by local Artin length. Explicitly, our nonstandard statistic is

$$N^{\mathrm{len}}(r) = \sum_{x\in X(\mathbb{F}_{p^r})} \operatorname{length}(\mathcal{O}_{X_{\overline{\mathbb{F}}_p},x}).$$

An orbit of degree d and geometric local length m contributes $(1-u^d)^{-m}$, so the weighted series is still a product of root-of-unity cycle factors with integer exponents.

Over characteristic zero, a reduced finite algebra decomposes as $\prod_i K_i$. Away from finitely many primes, the product of $Z_{a,p,n}(p^{-s})$ is therefore $\prod_i \zeta_{K_i}(s)$. This is a canonical Dedekind, equivalently permutation-Artin, interpretation. Such a global product can certainly have nontrivial zeros and can contain a Riemann-zeta factor. For example,

$$\zeta_{\mathbb{Q}(\sqrt{7})}(s) = \zeta(s)L(s,\chi_{28}), \qquad \zeta_{\mathbb{Q}(\sqrt{3})}(s) = \zeta(s)L(s,\chi_{12}).$$

These invariant-line factors are classical arithmetic components accompanied by extra Artin data. The obstruction is therefore not the absence of global zeros; it is that local rationality supplies no distinguished Hénon-derived Riemann divisor.

4.1 What the rectangular table forgets

We use arithmetic Frobenius $F_p : x \mapsto x^p$; geometric Frobenius reverses the sign convention in the second index. Because H_a is defined over $\mathbb{F}_p$, it commutes with Frobenius. The joint finite representation has the character

$$T_{a,p,n}(r,s) = \operatorname{Tr}\big(\mathrm{Frob}_p^r\, H_a^{-s} \mid \mathbb{Q}_\ell[S_{a,p,n}]\big) = \#\{x : \mathrm{Frob}_p^r\, x = H_a^s x\}. \tag{7}$$

The original data retain only $T(r,0)$.

Proposition 4.2 (Information loss)**.** *The sequence $r \mapsto N(r,n)$ does not determine the joint action of Frobenius and chronological Hénon time.*

Proof. On $\{\pm 1\} \times \mathbb{Z}/5\mathbb{Z}$, define

$$H(\varepsilon, i) = (\varepsilon, i+1), \quad R(\varepsilon, i) = (-\varepsilon, -i), \quad F_c(\varepsilon, i) = (\varepsilon, i + \varepsilon c).$$

Then $RHR = H^{-1}$, and each F_c commutes with both H and R. The choices $c = 1, 2$ both consist of two five-cycles, so $\mathrm{Tr}(F_1^r) = \mathrm{Tr}(F_2^r) = 10$ when $5 \mid r$ and zero otherwise. However, $\mathrm{Tr}(F_1 H^{-1}) = 5$ whereas $\mathrm{Tr}(F_2 H^{-1}) = 0$. $\square$

Thus a faithful two-axis continuation must retain twisted traces or all finite-index subgroup counts, not only rectangular fixed sets. Twisted counts of the form $\#\,\mathrm{Fix}(gF_p^r)$ already occur in equivariant finite-field zeta constructions [11, Definition 4.6]; the refinement is structurally appropriate but not a new construction type. Proposition 4.2 is a control in the category of reversible finite actions, not a claim that the concrete $a = 6, n = 5$ fiber realizes both alternatives.

5 Exact certificate and prior-work collision

All computations use exact integers, rational functions, and explicit finite fields. No target zeros, prime tables, or floating-point roots enter the protocol.

5.1 The first two periods at parameter six

At $a = 6$ the characteristic-zero algebras split as

$$X_{6,1} = \mathrm{Spec}\, \mathbb{Q}(\sqrt{7}), \qquad X_{6,2} = \mathrm{Spec}\big(\mathbb{Q}(\sqrt{7}) \times \mathbb{Q}(\sqrt{3})\big).$$

The second factor is the primitive period-two pair. At primes outside $\{2, 3, 7\}$,

$$N_{6,p}(r, 1) = 1 + \left(\frac{7}{p}\right)^r, \tag{8}$$

$$N_{6,p}(r, 2) = 2 + \left(\frac{7}{p}\right)^r + \left(\frac{3}{p}\right)^r. \tag{9}$$

Table 1 reports the complete frozen support-count rows. An independent polynomial-quotient implementation of every $\mathbb{F}_{p^r}$ enumerates solutions directly.

Table 1: Ordinary support counts for $r = 1, 2, 3, 4$. Weighted counts are kept in a separate ledger and coincide only in the ètale rows.

p	status	$N(r,1)$	$N(r,2)$
5	ètale-good	$0, 2, 0, 2$	$0, 4, 0, 4$
11	ètale-good	$0, 2, 0, 2$	$2, 4, 2, 4$
7	nonreduced	$1, 1, 1, 1$	$1, 3, 1, 3$
3	degree-drop	$1, 1, 1, 1$	$1, 1, 1, 1$

At $p = 7$, the local-length-weighted rows are $(2, 2, 2, 2)$ and $(2, 4, 2, 4)$; ordinary counts discard the double fixed-point length. At $p = 3$, direct reduction of the original uninverted family gives $H_0^4 = I$, so the $n = 4$ counts are $9, 81, 729, 6561$. This is not a fiber of the finite-flat family over $\mathbb{Z}[A, A^{-1}]$; mixing it into the uniform quadratic family would create a spurious large signal.

5.2 A tempting period-five signal

Restricting $\mathrm{Fix}(H_a^5)$ to the reversor line $q = p$ and removing the fixed branch gives a generic sextic $G_a(q)$ recorded in Appendix A.2. At $a = 6$ it is

$$46656q^6 + 15552q^5 - 20736q^4 - 4752q^3 + 3060q^2 + 360q - 151. \tag{10}$$

Its discriminant is $2^{36}3^{30} \cdot 31 \cdot 241 \cdot 389$. The modular factor degrees at the unramified primes 37, 5, and 157 are

$$[6], \qquad [5, 1], \qquad [2, 1, 1, 1, 1].$$

They give a transitive subgroup of S_6 containing a 5-cycle and a transposition. A 5-cycle excludes nontrivial blocks in degree six; conjugates of the transposition then generate S_6.

This exact result initially looks like a useful Galois-theoretic refinement. The rescaling $x = 6q$, however, turns (10) into

$$x^6 + 2x^5 - 16x^4 - 22x^3 + 85x^2 + 60x - 151.$$

Endler–Gallas already publish precisely this period-five Z polynomial, its discriminant $2^6 \cdot 31 \cdot 241 \cdot 389$, and its symmetric Galois group [4]. The scaling law

$$\mathrm{Disc}(Z(6q)) = 6^{30}\,\mathrm{Disc}(Z)$$

explains our powers $2^{36}3^{30}$. Brison–Gallas later republish the polynomial and supply companion sextics and polynomial bridges [1]. Equality holds in all seven coefficients. We therefore classify the calculation as a successful reproduction and a failed novelty gate no later than 2006.

5.3 Reproducibility summary

Eight unit tests pass. The independent checker verifies all 36 data cells, all 16 frozen field constructions, the period-five recurrence derivation, and the expected-fail joint-action control. Exact artifact hashes and commands appear in Appendix A.3.

6 Route-A evaluation and residual question

The local determinant in Theorem 4.1 is exact. Its exactness does not make it a candidate Riemann determinant: the determinant is universal for finite schemes and has no target divisor. Table 2 records the resulting Route-A decision.

Table 2: Scoped Route-A evaluation.

layer	verdict	decisive reason
A1	`A1_WEAK`	intrinsic cycles, but no prime-like clock or amplitude
A2	`A2_FAIL`	universal root-of-unity permutation determinant
A3	`A3_FAIL`	no completed Riemann divisor or counting law
A4	`A4_FAIL`	no natural operator lift of the arithmetic scheme factor

The overall label is `ROUTE_A_REJECTED`; Route B is not authorized for this registered candidate. The failure is structural rather than numerical. This ruling does not deny the nontrivial zeros of the classical global Dedekind/Artin factors, nor does it close every joint or positive-dimensional mechanism. It says that the fixed-n local rationality and finite trace decomposition are universal controls and hence have no discriminating value for a new Hénon-derived Riemann divisor.

A narrower arithmetic problem survives. On exact-period points, absolute Galois commutes with H_a and with the reversor R. Its image therefore lies in the centralizer of a dihedral action, which is generally smaller than the cyclic wreath-product bound. Low-period polynomial and Galois calculations are already well represented in the literature, so a new result would need to determine a genuinely new higher-period image, a uniform proper-subgroup obstruction, or the cohomology of a parameter-varying quotient. None of these claims follows from fixed-n rationality.

7 Conclusion

The area-preserving Hénon recurrence has a clean arithmetic periodic scheme: away from coefficient degree-drop primes, its n-th fixed scheme is finite flat of rank 2^n. That clean structure also closes the registered local-zeta route. At fixed n, Frobenius acts on a zero-dimensional support, so local rationality and finite trace reconstruction are inevitable finite-permutation facts. Nilpotents disappear from ordinary counts, and rectangular counts do not retain relative chronological phase.

The obstruction is one of novelty, not a theorem forbidding global zeros. Fixed finite algebras globalize to classical Dedekind/Artin zeta functions, and may inherit a Riemann-zeta factor together with additional L-data. Nothing here rules out a genuinely new joint or positive-dimensional construction.

The exact certificate matters mainly as a firewall. It verifies good, ramified, and degree-drop conventions and prevents a known period-five S_6 sextic from being misreported as a new arithmetic signal. Future work should not enlarge the same r-scan. A defensible next candidate must introduce new geometry—for example a positive-dimensional exact-period parameter quotient—or prove a new theorem about the full dihedral-compatible Galois tower. Neither direction currently supplies a Hilbert–Pólya operator.

References

[1] Owen J. Brison and Jason A. C. Gallas. Polynomial interpolation as detector of orbital equation equivalence. *International Journal of Modern Physics C*, 29(8):1850096, 2018. doi: 10.1142/S0129183118500961.

[2] Jeffrey Diller and Charles Favre. Dynamics of bimeromorphic maps of surfaces. *American Journal of Mathematics*, 123(6):1135–1169, 2001. doi: 10.1353/ajm.2001.0038.

[3] Antônio Endler and Jason A. C. Gallas. Arithmetical signatures of the dynamics of the hénon map. *Physical Review E*, 65:036231, 2002. doi: 10.1103/PhysRevE.65.036231.

[4] Antônio Endler and Jason A. C. Gallas. Reductions and simplifications of orbital sums in a hamiltonian repeller. *Physics Letters A*, 352(1–2):124–128, 2006. doi: 10.1016/j.physleta.2006.01.031. URL https://inaesp.org/PublicJG/endler_gallas_orbital_sums_PLA2006.pdf.

[5] Shmuel Friedland and John Milnor. Dynamical properties of plane polynomial automorphisms. *Ergodic Theory and Dynamical Systems*, 9(1):67–99, 1989. doi: 10.1017/S014338570000482X.

[6] Alexander Grothendieck. Formule de lefschetz et rationalité des fonctions L. In *Séminaire Bourbaki, Vol. 9, Exp. No. 279*, pages 41–55. Société Mathématique de France, 1964–1966. URL https://www.numdam.org/article/SB_1964-1966__9__41_0.pdf.

[7] Michel Hénon. Numerical study of quadratic area-preserving mappings. *Quarterly of Applied Mathematics*, 27:291–312, 1969. doi: 10.1090/qam/253008.

[8] Benjamin Hutz. Dynatomic cycles for morphisms of projective varieties. *New York Journal of Mathematics*, 16:125–159, 2010. URL https://nyjm.albany.edu/j/2010/16-8p.pdf.

[9] Douglas A. Lind. A zeta function for $\mathbb{Z}^d$-actions. In Mark Pollicott and Klaus Schmidt, editors, *Ergodic Theory and $\mathbb{Z}^d$-Actions*, volume 228 of *London Mathematical Society Lecture Note Series*, pages 433–450. Cambridge University Press, 1996. doi: 10.1017/CBO9780511662812.019. URL https://sites.math.washington.edu/~lind/Papers/ZetaFunction.pdf.

[10] John A. G. Roberts and Franco Vivaldi. Signature of time-reversal symmetry in polynomial automorphisms over finite fields. *Nonlinearity*, 18(5):2171–2192, 2005. doi: 10.1088/0951-7715/18/5/015.

[11] Laura Walton. Counting periodic points on quotient varieties over $\mathbb{F}_q$. *Journal of Number Theory*, 192:386–405, 2018. doi: 10.1016/j.jnt.2018.03.023. URL https://arxiv.org/abs/1705.09034.

A Proof details and reproducibility

A.1 The low-period normalized Jacobian norms

For $n = 1$, set $f = ax^2 + 2x - 1$ and $j = 2ax + 2$. Taking the norm of j in $\mathbb{Q}(a)[x]/(f)$ gives

$$\mathrm{Norm}(j) = -4(a + 1).$$

For $n = 2$, the cyclic Jacobian determinant is

$$j_2 = 4a^2x_0x_1 - 4.$$

Multiplication by j_2 on the standard basis $1, x_0, x_1, x_0x_1$ has determinant

$$2^8(a + 1)(a - 3)^3.$$

These $D_{a,n}$ are normalized transversality resultants/Jacobian norms, not ordinary polynomial discriminants (the ordinary $n = 1$ quadratic discriminant is $4(a + 1)$). The factorization

$$f_0 - f_1 = (x_0 - x_1)(a(x_0 + x_1) - 2)$$

separates the fixed branch from the primitive period-two branch. With $u = ax+1$ and $v = ax_0-1$ the two quadratic relations are

$$u^2 = a + 1, \qquad v^2 = a - 3.$$

The branch ideals are comaximal over $\mathbb{Q}(a)$, so this gives the scheme-theoretic product there. At $a = 3$ the primitive branch collides with the fixed branch; the splitting is not asserted at that specialization.

A.2 The generic reversor-line marker

The exact period-five marker before specialization is

$$\begin{aligned} G_a(q) = {} & a^6q^6 + 2a^5q^5 + (-3a^5 + 2a^4)q^4 + (-4a^4 + 2a^3)q^3 \\ & + (3a^4 - 4a^3 + a^2)q^2 + (2a^3 - 2a^2)q - a^3 + 2a^2 - a - 1. \end{aligned}$$

It is obtained as the greatest common divisor of the two equations defining $\mathrm{Fix}(H_a^5)$ after setting $p = q$, divided by the fixed-point factor $aq^2 + 2q - 1$. The producer derives this expression; it is not inserted as a fit.

A.3 Reproducibility ledger

From the repository root, run

```
python henon_frobenius_scheme_obstruction/code/test_c12a.py
python henon_frobenius_scheme_obstruction/code/c12a_producer.py
python henon_frobenius_scheme_obstruction/code/c12a_checker.py
```

The frozen artifacts have SHA-256 values

artifact	SHA-256
certificate JSON	851ca31f62fb508ad806c26084eab9fe092d5ee037bf99f0cb811cbccf7f8eb8
count CSV	d07d9558dd9036507b89699452edc1494bea2faca8344d5ce6cf2d031f9bc480
independent check	4784e8b2fbf98ad835a5f1c0ef9217de14537adcff486046e74a6b0f47e93778

The checker does not import the producer. It constructs each frozen finite field as an explicit polynomial quotient, verifies directly that every nonzero element is a unit, and enumerates the cyclic equations. The period-five checker starts instead from the scalar recurrence on the reversor line. It also enumerates all states of the reversible joint-action control and verifies $RHR = H^{-1}$ and both commutation relations. The v2 amendment that introduced this control is recorded in `AMENDMENT_LOG.md`.

A.4 Limitations

We do not prove that $X_{a,n}$ is reduced in characteristic zero for every fixed a and n. Scheme-level formal period requires extra care when the characteristic divides n. The global Dedekind factors discussed in the main text can have nontrivial zeros and may contain $\zeta(s)$; what they do not provide is a new Hénon-specific gamma factor, exact Riemann divisor, or self-adjoint realization required by Hilbert–Pólya.